
Must Success Make Sense?

Jaia Mridula^{1,2} and David Elks¹,

¹ Data and Analytics Function, UK Government Investments (UKGI)

² Department of Science, Technology, Engineering and Public Policy (STePP),
University College London (UCL)

Abstract: Explainable AI begins from a curious, powerful hope. If the operations of an artificial system can be explained, then its opacity can be overcome. This paper questions that hope. Advanced AI systems increasingly produce answers, predictions and what some consider successful discoveries, insofar that this success can be recognised even when the reasoning that produced them evades human graspability. In these cases, explanation may inform us without restoring understanding. The paper argues that intelligibility is a distinct epistemic achievement. To understand is to not receive an account of a result, it is to be able to inhabit the relations that make the result meaningful. Turning to the philosophy of science and of mathematics, the paper turns to mathematical beauty as an image of achieved understanding. A beautiful proof does more than establish truth by making that truth mentally possessable. We draw upon cases from AlphaFold, large language models and AI mathematical reasoning to show why this distinction is a timely, contemporary matter. These systems suggest a future in which human beings can rely on successful reasoning that they can no longer make fully their own. This would suggest the task of human centred AI is therefore not only to explain machines but to defend the place of understanding in a world shaped by their success.

Keywords: artificial intelligence, explainability, intelligibility, opacity, understanding

1. Introduction

Nowadays, an artificial system can give you an answer that no human 'expert' could have reached with the same speed or range. It predicts a protein structure, supplies a line of mathematical reasoning and produces an account that appears to justify its conclusion of both [1–4]. That result can be tested, compared and perhaps even accepted amongst its academic community. But beyond recognition of its success, what has actually become intelligible to us? This question marks a departure in the philosophical problem posed by artificial intelligence. Earlier anxieties about computation were built around error or the delegation of judgement but these contemporary systems are emblematic of a different concern entirely. They can succeed in domains whose value has traditionally depended not only upon reaching results but upon understanding why those results hold [1, 3–5]. Humphreys [6] describes epistemic opacity as a condition in which computational processes contribute to knowledge while exceeding direct human survey. Explainable AI has responded to this condition by seeking forms of access to machine behaviour through transparency, as well as interpretation and explanation [7–10]. There is little contention that the systems that stand to affect human lives or participate in inquiry must be open to criticism and responsible use, yet it remains possible that a reading of a machine's output will leave the meaning of its achievement outside human grasp.

Our paper asks, **can human understanding remain an epistemic requirement when advanced AI systems succeed through processes that exceed human cognitive access?** Our answer is that human understanding can and should remain such a requirement but only if it is differentiated from explanation alone. Explainability addresses an important problem of access because it asks how an output

can be inspected and used, and by extension, defended, by a human agent. Understanding names a further achievement. It concerns the capacity to grasp relations that make a result meaningful and to carry that grasp into criticism and inquiry. Through de Regt’s [11,12] account of intelligibility in scientific understanding, this paper argues that the deepest challenge of advanced AI is that successful reasoning can increasingly occur in forms that humans can employ without being able to inhabit. It develops this claim through the philosophy of science and the philosophy of mathematics, where the distinction between a correct result and a profound one has long been expressed through accounts of explanation and mathematical beauty [5,13,14]. It then tests the argument through cases like AlphaFold, large language models and AI mathematical reasoning [1–4,15]. These cases show why the preservation of understanding can no longer be assumed only because a system performs well or offers an explanation.

2. Approach

We use an approach that places explainable AI in conversation with the philosophy of scientific understanding which depends upon intelligibility and the skills through which scientists can use a theory to grasp phenomena [11,12]. We extend that question to AI to ask whether machine success can support human understanding when the forms of reasoning responsible for that success no longer become cognitively available to us. The argument develops through three movements. It first reconstructs explainable AI on its own terms by considering how transparency, interpretation, explanation and trust are invoked as responses to opaque machine learning [7–10]. Miller [8] explicitly argues that explanation in AI must be informed by the social and cognitive practices through which people seek reasons from one another. Zednik [10] similarly proposes a normative framework in which different stakeholders have different explanatory questions. These contributions make the present argument possible since they show that explanation is never simply the extraction of internal information. The paper then asks whether even a well designed explanation must produce understanding. To develop this distinction, it turns to the philosophy of mathematical practice where the difference between establishing a result and seeing its significance has long been marked by the language of elegance and beauty [5,14,16,17]. Finally, that distinction is tested through three cases. AlphaFold concerns prediction within a science that seeks causal and structure grasp [1,18]. Large language models concern fluent rationales whose relation to underlying computation remains uncertain [15,19,20]. AI mathematical reasoning concerns proof, where correctness can be secured while insight is absent [3–5]. These all allow us to claim that systems used in inquiry, discovery and proof place pressure on a human epistemic value that cannot be measured by successful output alone.

3. The Promise of XAI

Explainable AI (XAI) develops from a familiar experience of modern machine learning when a system seemingly predicts or recommends with striking success while those who build it or are customers of it remain unable to give a satisfying account of a specific result as the output [7–9]. Lipton [7] warns the language of interpretability gathers together disparate ambitions and a model can be: transparent in its design; simple enough to simulate; decomposable into individual parts; or accompanied by a post hoc account of a particular output. But these are not all the same achievement. Doshi-Velez and Kim [21] likewise argue that interpretability requires a more rigorous science because its aims and measures change with the task and the intended user. Therefore, whilst we are already familiar with the problem of some systems as black boxes, the challenge we confront going forward is that the demand to open the box conceals several different human needs. Arrieta et al. [9] cover this by describing XAI as central to responsible AI and by organising methods around the audience and purpose for which an explanation is sought. On this view, opacity matters because it weakens the relation between machine output and human judgement. Explainability promises to repair that relation by returning reasons or representations of scrutiny to the people whose actions and knowledge increasingly depend upon artificial systems [8–10].

The force of that promise is explicated once its main concepts are kept apart. Transparency concerns some form of access to the system, whether to its architecture, variables, training procedure or internal operation [9]. Interpretability concerns the extent to which what is disclosed can be made revelatory for a pertinent human purpose [21]. Explainability concerns the production of an account that addresses a question

about an output or behaviour [8, 10]. Trust concerns the willingness, and sometimes the justification, to rely upon the system in practice [22]. Much XAI research works across these concepts instead of being assigned in a silo manner to a single fixed order. For example, Rudin [23] argues that in high stakes settings an innately interpretable model should be preferred to a black box model followed by a post hoc explanation. The objection is to the risk that an explanation of an opaque model can provide reassurance without fidelity and not to explanation itself. Mittelstadt, Russell and Wachter [24] make a related point from a different perspective. Simplified models can be useful in facilitating a practitioner to explore what a system might do but this usefulness should not be confused with a complete philosophical account of explanation. Miller [8] adds that human explanations are selective and contrastive. They are also social because an explanation answers a question for someone and within a context. These arguments show that XAI already goes further than the simple ideal of exposing the technicalities of every internal operation, and seeks out forms of access that are relevant to evaluation and accountability [24].

Our argument begins where the more reflective account of XAI dwindles down. If explanations are addressed to particular agents and purposes, then their success cannot be identified with the quantity of information disclosed. Full transparency is perhaps impossible for a complex system, and even where it is possible it may give a human reader only an unmanageable record of computation. A feature attribution or a counterfactual example can be more useful than such a record because it selects what matters for a question. Albeit selection brings a further difficulty. An account can be purposeful for use while leaving the significance of the system’s achievement untouched. Sokol and Flach [25] move close to this issue when they define explainability as a process that can engender understanding for an explainee with background knowledge. Zednik [10] likewise connects explanatory demands to stakeholder purposes, and the question developed from this connection is around what happens when no available explanation makes the successful reasoning humanely inhabitable, even though outputs remain usable and reliable. This would suggest intelligibility is the limit and if so, the promise of XAI must then be measured against an epistemic achievement deeper than explanation itself.

4. Explanation is Not Understanding

An explanation is something that can be offered. Understanding is something that must be achieved. This difference is easy to overlook because explanation is commonly sought in the hope of understanding, and a good explanation will often bring understanding about. This means their identification from one another is quite difficult. In epistemology, Kvanvig [26] argues that understanding has a value that is not exhausted by the acquisition of isolated true beliefs. Grimm [27] similarly treats understanding as involving a grasp of dependence or connection and not a summative accumulation of truths. Elgin [28] develops this point by showing understanding can involve models and abstractions whose epistemic value lies in the way they facilitate a subject to organise and reason about a domain. These accounts are important for AI because a system can generate true statements without giving human agents this grasp. If a human subject cannot place that information within a structure of reasons, that result cannot be made feasibly profound. Explanation takes a step toward that movement but cannot alone guarantee its completion.

Philosophy of science gives this distinction a form directly suited to the problem of advanced AI. De Regt and Dieks [11] argue that scientific understanding depends upon intelligibility, which is achieved when scientists possess the skills needed to use a theory in the construction of explanations. De Regt [12] develops this contextual account by resisting the idea that understanding requires a single preferred kind of representation. For example, visualisation and idealisation can each make phenomena intelligible when they allow competent agents to reason with them. The argument of this paper is not to liken human cognition to intelligibility. Rather, we take it to mean that the achievement of a system can be taken up within human practices of sense making like inquiry and criticism. Khalifa [29] offers a different account which ties scientific understanding more closely to explanatory knowledge and reinforces the point that understanding must concern the explanatory relation more than it does the possession of a bare answer. For advanced AI, this gives rise to an interesting proposition that trust is not the endpoint. We may trust a tool because it has performed reliably against the task it was commissioned. Understanding asks whether its success can become part of what we ourselves are able to see and reason through.

And so, the difference between explanation, understanding and its constituent parts can be stated even

more precisely. None of the facets of performance, correctness, transparency, explainability, or interpretability are negligible, nor do they alone secure intelligibility. Intelligibility concerns whether the relations on which a result depends can enter human thought in a form that supports grasp and further inquiry. Understanding is the achievement made possible when this condition is met. In that regard, explanation is an epistemic offer, while understanding is an epistemic possession. An explanation that supports a person in contesting an automated decision may be entirely successful for that purpose. An explanation offered in science or mathematics faces an even larger burden because the value of the output is bound to whether it improves human insight. At this point, we turn to mathematics. This is because mathematical practice has long recognised that a proof can be valid without being profound and that a solution can command assent without yielding the experience of seeing why it is true. The language of mathematical beauty gives historical expression to this difference and also gives us a way to prod further what is lost when AI produces correct and powerful results whose success cannot be made our own.

5. Mathematical Beauty

Mathematics provides an especially clear background in which correctness and intelligibility come apart. Mathematical practice has never valued demonstration alone and has also valued forms of reasoning that disclose structure and gather complexity into relations that can be grasped. Poincaré [16] gives this connection a significant early expression in 'Science and Method' where he argues that mathematical invention does not arise from mechanical combination alone. Indeed, the mathematician must recognise which combinations are fruitful, harmonious and capable of revealing an order otherwise concealed within a mass of possibilities. Beauty, in this account, concerns the recognition of significant relation which matters for our argument because an advanced artificial system can search a vastly larger field of possible combinations than any human mind can [3, 4]. Through that exercise, it may identify a pattern and prove its worth by successful output but the human recipient lacks the form of insight through which that pattern gains any meaning. Poincaré [16] hence helps to frame the problem we confront as whether the movement from possibility to purposeful structure can still become available to human understanding when selection has been made by a system whose cognitive path we cannot follow.

In a slightly different register, Hardy [17] and Weyl [30] express the same concern. 'A Mathematician's Apology' sees Hardy associate mathematical beauty with economy, unexpectedness and inevitability [17]. A significant proof bears a form in which the mind can recognise that the result could not have been otherwise, even where the path to it was surprising. Weyl [30] similarly treats mathematical form as animated by an aspiration towards order and beauty, even while acknowledging that truth remains the final obligation of mathematical work. These two accounts develop a specific relation between form and comprehension. Rota [14] builds that out more explicitly when he describes mathematical beauty through the experience of enlightenment. A proof becomes beautiful where it makes an initially obscure result appear newly evident through an organising idea and so, its value is in the transformation it effects within the mind of the reader [14]. This gives mathematical beauty its relevance to AI because if no organising idea becomes visible through the machine's achievement of having delivered an answer, then what has been gained remains confined external to human thought. We draw upon beauty here because it names one historical form of the experience in which correctness ceases to be accepted and becomes intellectually possessed [5, 14].

We need not rely on aesthetic testimony to make the philosophical case. Steiner [5] distinguishes explanatory proofs from proofs that establish a mathematical result because according to him, an explanatory proof reveals why a theorem holds by drawing upon a property that characterises the relevant entities within the mathematical domain. Kitcher [13] approaches explanation through the unifying power of argument patterns. Mathematical understanding increases when apparently separate results can be derived through a small and graspable family of reasoning forms. These accounts diverge slightly but both reject the idea that proof is exhausted by valid derivation. They make room for a further achievement in which a theorem is placed within a structure of dependence or unity but this is what becomes uncertain in the presence of advanced AI. An artificial system can construct a valid proof that extends the known mathematical record and may go so far as to render that record in readable notation. This readability is not identical to explanation, and explanation is not identical to insight. Therefore, the appeal to mathematical beauty should be understood with restraint as a philosophical witness to a distinction already present within human reasoning.

6. Examples of Success

We draw upon three examples in this section through which successful output and intelligible achievement can be explored. The first is AlphaFold, which Jumper et al. [1] describe as a system capable of predicting protein structures with accuracy competitive with experimentally determined structures in a majority of cases in the CASP14 assessment. As a scientific accomplishment, this is extraordinary because protein structure bears upon biological function, and reliable structural predictions can predict research that would otherwise require long and difficult experimental work. However, AlphaFold predicts a structure from sequence and related information and it does not thereby supply a complete account of how a protein folds in a cellular environment or how its conformations shift through interaction. Terwilliger et al. [18] make this limit explicit because they find that AlphaFold predictions frequently reconcile closely with experimental maps, but also find departures in side chain conformation and interactions involving environmental factors not represented in the prediction. They therefore describe these predictions as exceptionally useful hypotheses which accelerate experimental structure determination [18]. The philosophical significance of AlphaFold is that it shows a machine generated result can transform inquiry without completing the movement to understanding and its predictions gain scientific traction through the further work of testing and causal investigation. Ergo, it shows that the relation between prediction and understanding must be kept open as a question.

Large language models create a different and in some respects more troubling form of distance between explanation and understanding because their outputs occur in the medium through which humans ordinarily exchange reasons. They can provide an answer and then accompany it with a sequence of sentences that appears to explain how the answer was reached. Wei et al. [2] showed that chain of thought prompting can improve performance on tasks involving arithmetic and symbolic reasoning. Such results make generated rationales attractive as a possible window to better answers and machine reasoning but this inference is relatively fragile. Mitchell and Krakauer [20] argue that debates about whether language models understand cannot be settled by pointing to their linguistic competence because the relevant question concerns the forms of relation and abstraction that their performance expresses. More directly, Turpin et al. [15] show that chain of thought explanations can systematically fail to reveal the factors that influence a model's prediction so a model can generate a persuasive rationale for an answer affected by a biasing feature without acknowledging that influence. Lanham et al. [19] likewise find that the faithfulness of stated reasoning varies across tasks and model scales. These results can still facilitate a user to explore an argument or identify an error but their limit is nevertheless epistemic. An articulate explanation can function as a useful object of reasoning for a human reader while remaining an uncertain account of the process that produced it, appearing to supposedly speak in the grammar of reasons but still not returning the reasoning directly into human possession.

All mathematical reasoning brings these concerns into the domain where the connection between correctness and insight has been most carefully explored. Trinh et al. [3] present AlphaGeometry as a neuro symbolic system that solved 25 of 30 translated Olympiad geometry problems and approached the performance of an average international Mathematical Olympiad gold medallist. AlphaGeometry generates proofs that can be read and evaluated by human experts, which shows that artificial mathematical achievement, one that takes the form of a readable proof, proceeds to leave open a question about insight. This question is refined by AlphaProof. Hubert et al. [4] describe a reinforcement learning system operating within Lean, where proof terms can be verified by a formal kernel. Together with AlphaGeometry 2, it achieved a score equivalent to a silver medallist at the 2024 International Mathematical Olympiad. The formal verification provides certainty that the constructed argument satisfies the rules of the formal system but this form of certainty does not itself establish that mathematicians understand the strategy discovered by the system, or that the proof reorganises a domain in an innovative manner. Returning to our discussion from the previous section, this shows that correctness can be mechanised but insight is still human.

7. Objections

A first objection is that dependence upon what we do not fully understand is neither new nor especially troubling. It would be injudicious to claim every individual understands every instrument or dataset on which contemporary science depends. Humphreys [6] already shows that computation science forces inquiry beyond what human agents can directly survey. Why, then, should advanced AI demand a new principle

of intelligibility? The answer is that understanding has always been distributed and mediated. However, the apprehension lies where systems cease to extend human inquiry and begin to occupy the place at which reasons and discoveries are generated. It is not a reason to reject the tools at our disposal but it is a reason to distinguish reliance from epistemic participation. A second objection follows naturally. Perhaps reliable performance is enough and in some settings, it is. But where AI participates in more complex fields like science, mathematics, medicine or public judgement, users should ask what a result reveals, when it fails and how it may be carried into new circumstances. Reliability permits action and intelligibility permits the growth of knowledge.

A further objection is that the problem is temporary. Mechanistic interpretability could one day plug the missing gap between artificial computation and human grasp. Rauker et al. [22] describe such work as a route towards understanding the inner structures of deep neural networks and improving their trustworthiness. This work matters deeply to our argument because a system whose mechanisms are better understood is more open to diagnosis and challenge. However, the disclosure of mechanism cannot by itself guarantee intelligibility and there is no guarantee that a complete recount of the activations or circuits are coupled tightly enough to the conceptual structure of a scientific or mathematical question to support understanding of the result. Mechanistic interpretation can serve intelligibility without exhausting it. A final objection concerns mathematical beauty. Beauty was referred to in this argument to show there is a visible distinction between result and insight, even though each of the philosophers we referred to expressed it in different terms. Artificial intelligence can both amplify the production of results and weaken the conditions under which results become profound for human minds. Intelligibility is a demand we assign to machines in fields concerned with knowledge and reason so that its successes can faithfully enter practices of human inquiry.

8. Conclusion

Human understanding can remain an epistemic requirement only where we refuse to identify knowledge with the availability of successful outputs. In lines and domains of inquiry, the value of intelligence is not in the production of correctness but in supporting humans in their interpretation of seeing what the right result means. Scientific and mathematical understanding have always depended upon instruments and forms of shared expertise but what must be preserved is the possibility that machine produced results can be integrated into human practices of reason as intelligible achievements. Explanation matters where it contributes to that possibility. If the machine can teach the human how to see and extend its findings, then its explanation and understanding are upheld in equal measure. Where it fails in that movement, explanation supports reliance while leaving understanding behind. The challenge for future research is hence to move beyond asking whether AI can generate explanations and to ask what kinds of explanation sustain human cognition within fields of discovery. Whether artificial intelligence should be placed beneath a human standard that cannot be exceeded is not a debate germane to this conversation. Instead, we seek to land that there is a difference between reaching a result and coming to see it. If advanced AI seeks to make the first achievement routine while leaving the second behind, then its success poses an epistemic problem to humankind. As machine intelligence expands the range of what can be known, human beings should not lose their place in understanding what that knowledge means.

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